Bookstore Analytics

A Data Engineering Solution for Bookstore Sales Analytics

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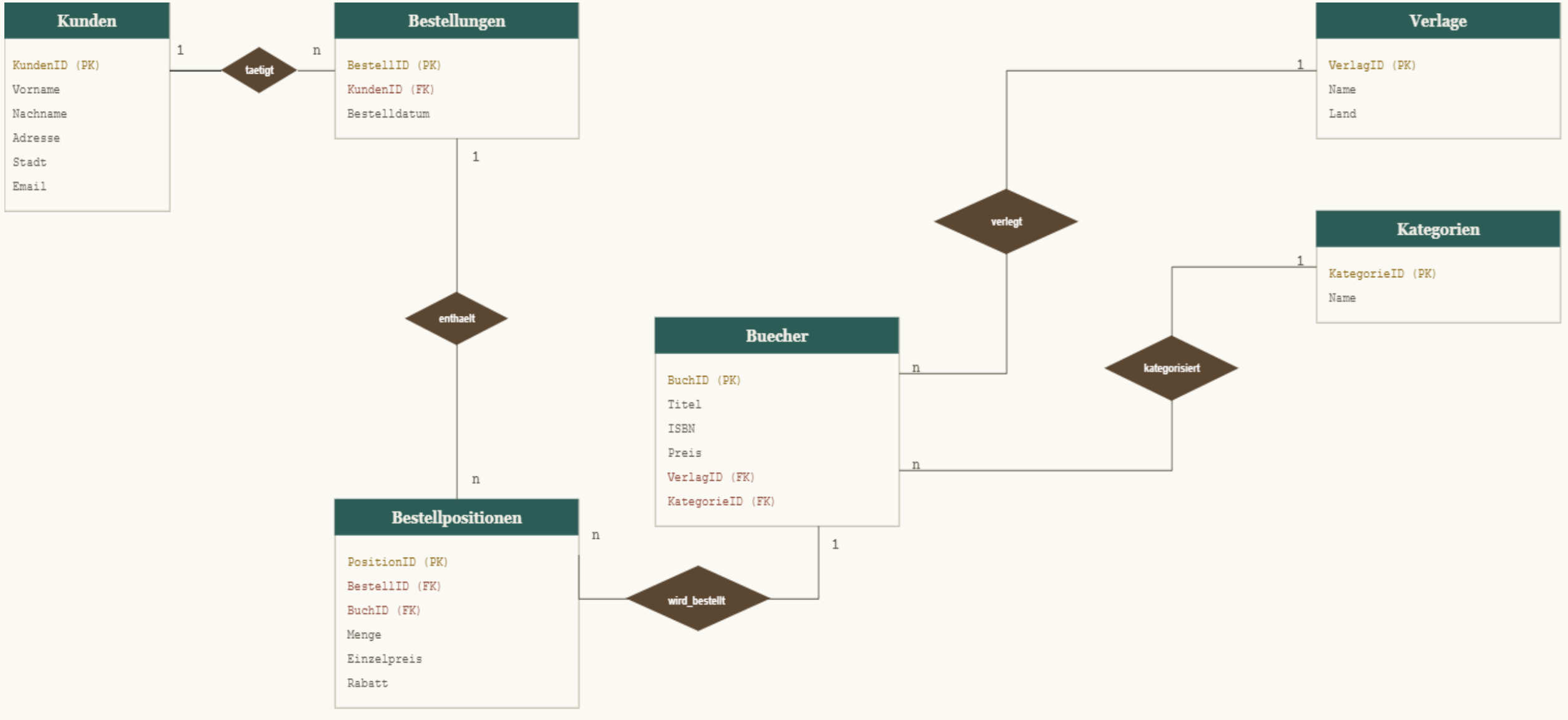
Project Goal & Scope

- Online bookstore ordering process
- Business DB → ETL → Data Warehouse
- Historical data management with SCD

Constraints

- Each book is assigned to exactly one publisher
- Foreign keys ensure that references between related tables are valid
- Each book must have a unique ISBN
- Important business attributes such as name, price, date, and quantity are mandatory
- If no discount is specified for an order item, the default discount is 0

Data Model (ERD) & Schema

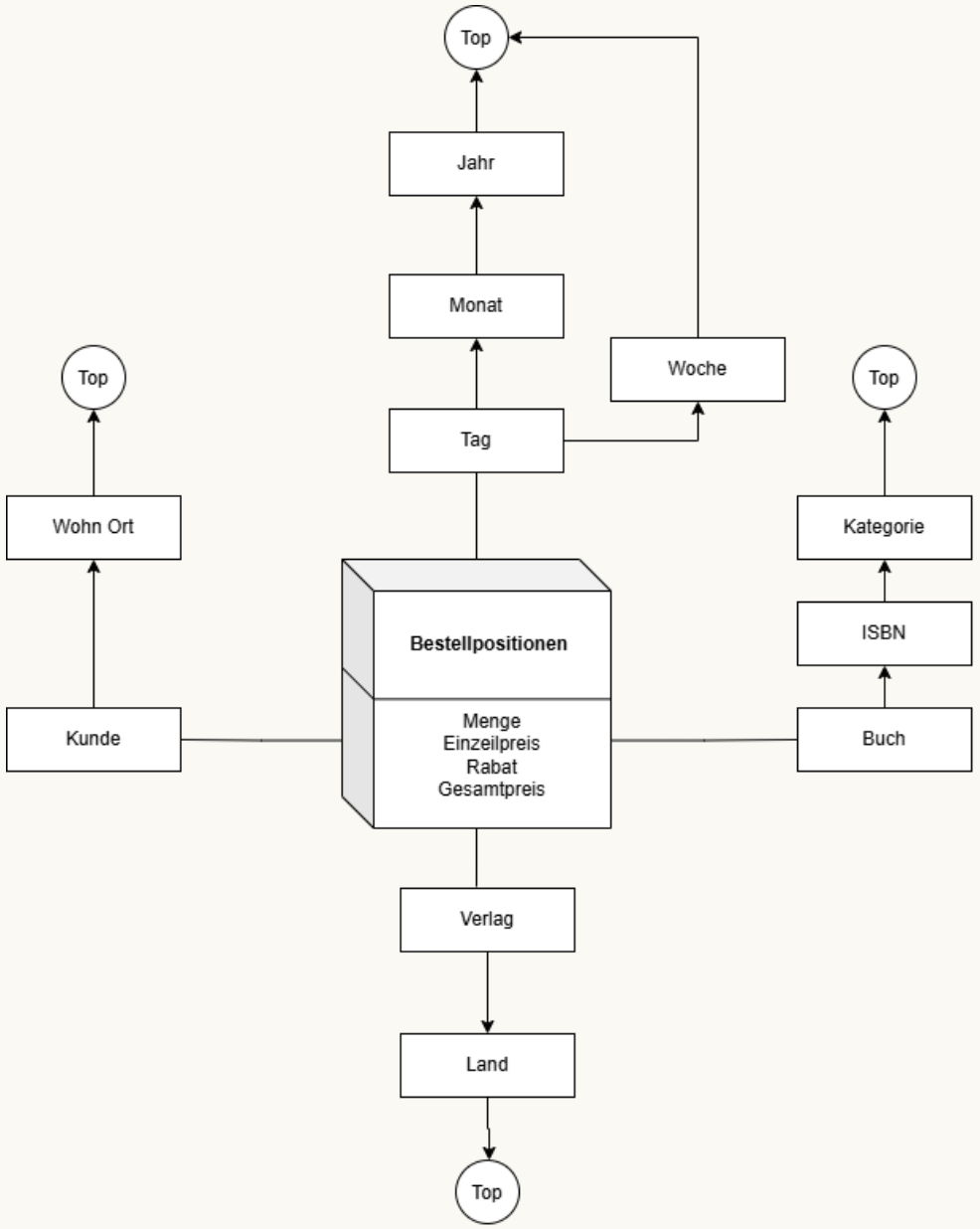


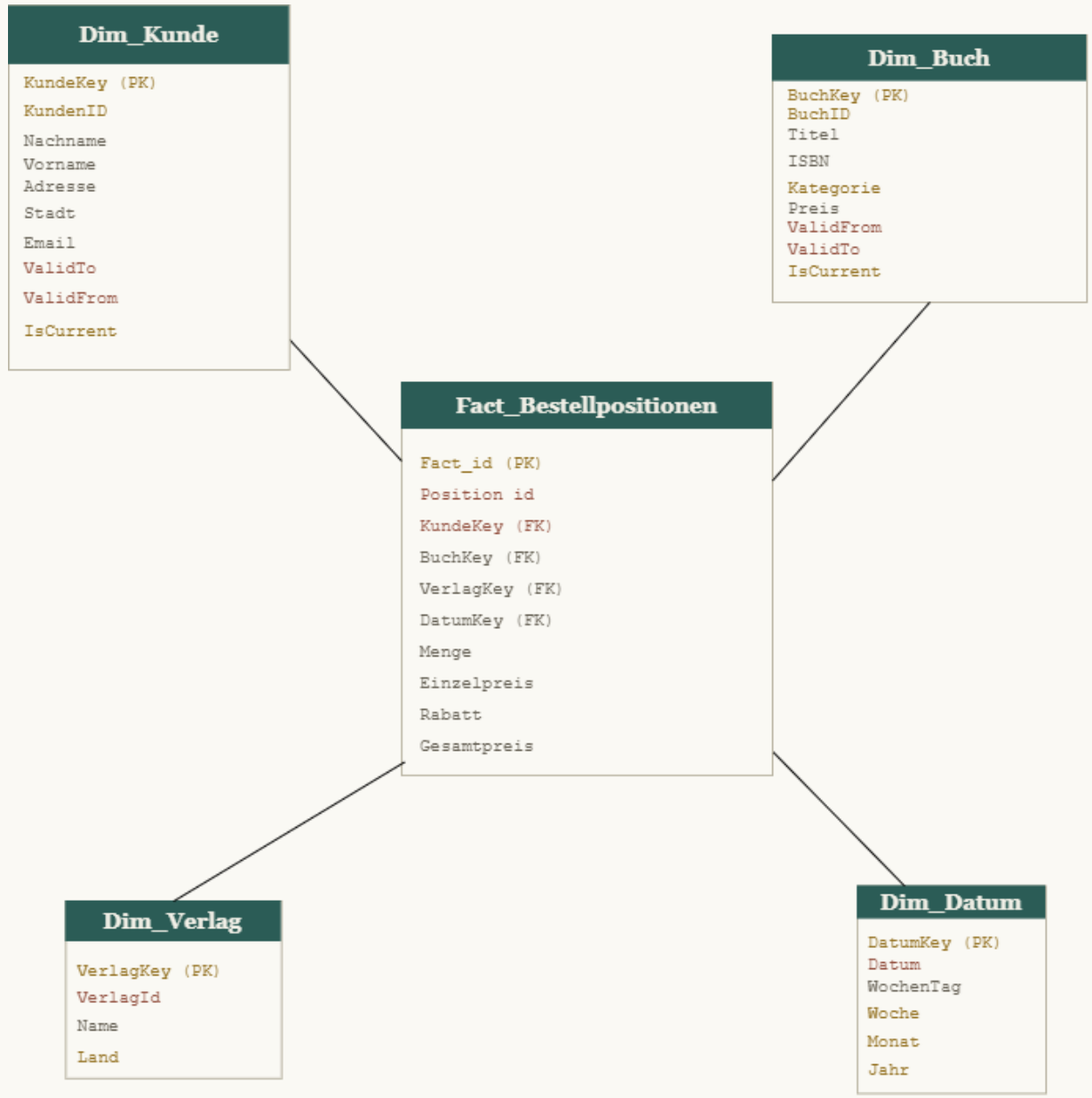
Data

- Covers the ordering process of a small online bookstore
- Order data covers about 3 weeks (Aug. 3–19)
- Uses manually created, fictional data
- Data provided as SQL INSERT statements
- Small dataset designed to be easy to understand and trace

Table	Number of rows	Description
Customers	6	Customer data
Publishers	5	Publisher data
Categories	4	Fiction, Crime, Non-fiction, Children's books
Books	16	Each assigned to 1 publisher and 1 category
Orders	25	Distributed over approx. 3 weeks
Order Items	35	Individual book items per order

DWH Design: mER Model + Star Schema





Setup: Two Databases

Bookstore Analytics - one operational database, one analytical database

BookstoreBusinessDB

Operational (OLTP)

- 6 tables, normalized to 3rd normal form
- Source of truth for live orders as they happen
- Optimized for fast, safe, single-row transactions

Python ETL

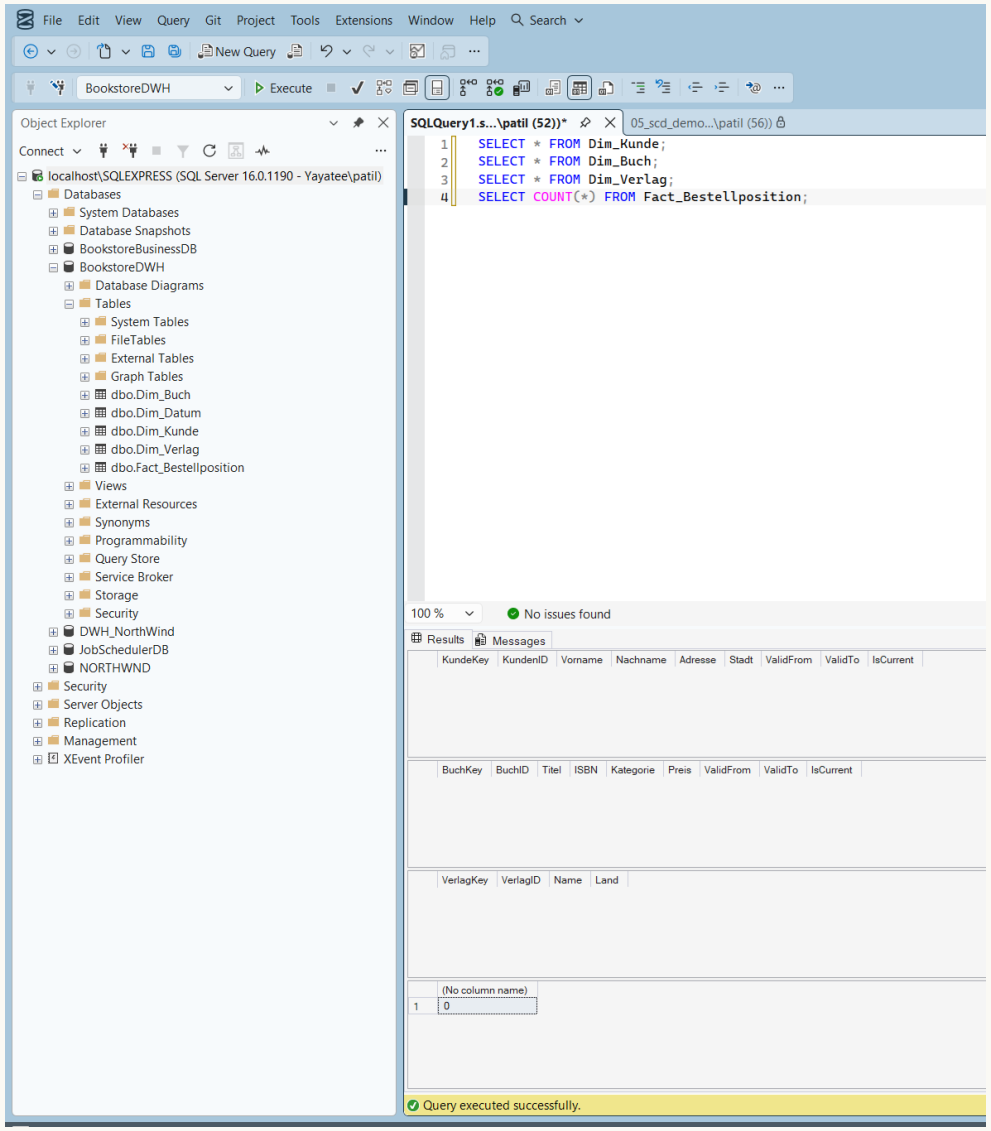
06_etl_mssql.py

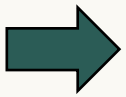
BookstoreDWH

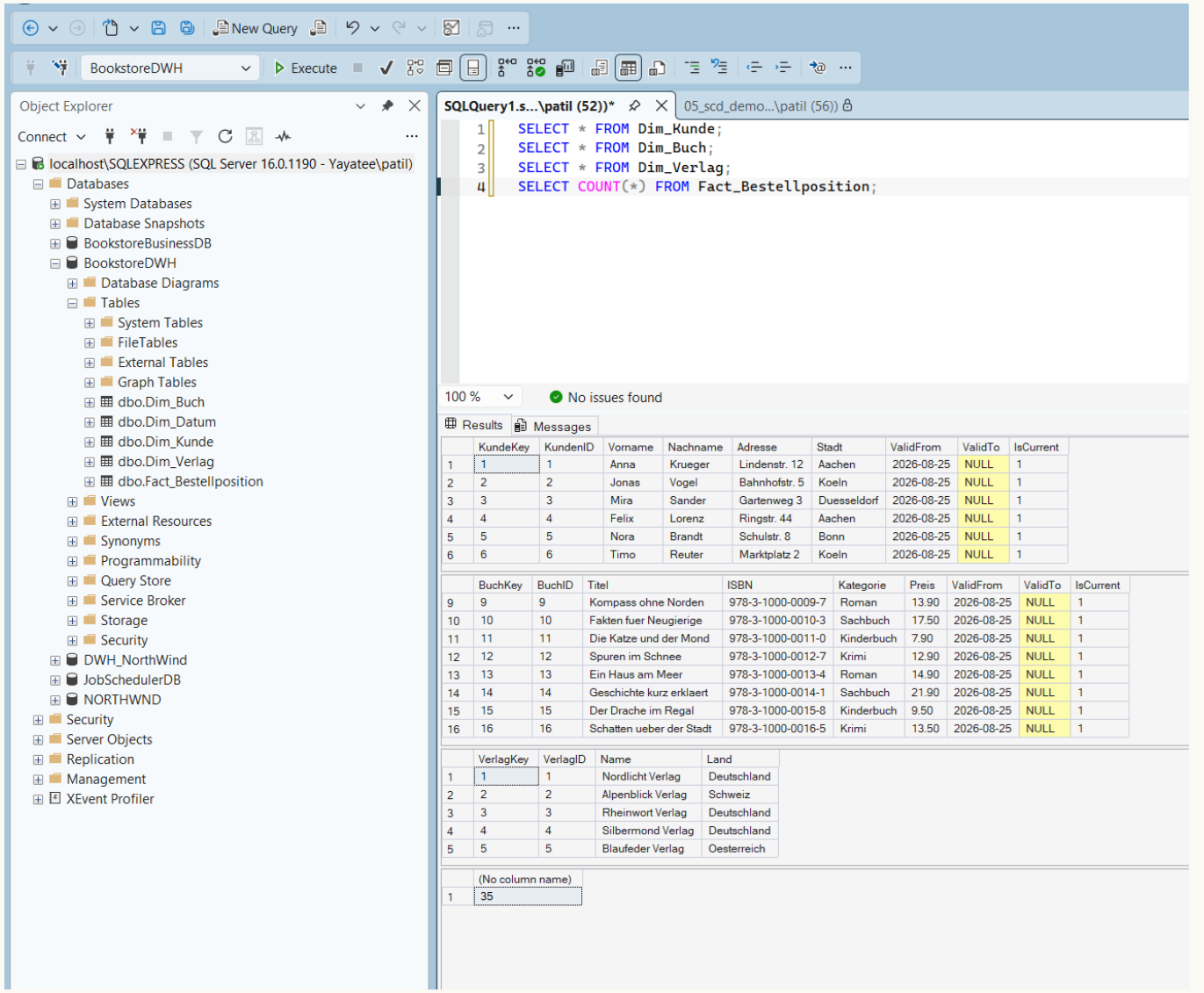
Analytical (Star Schema)

- 1 fact table + 4 dimensions
- Denormalized on purpose, for fast reporting
- Keeps history via SCD Typ 1 / Typ 2

First load in DWH





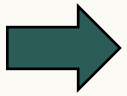


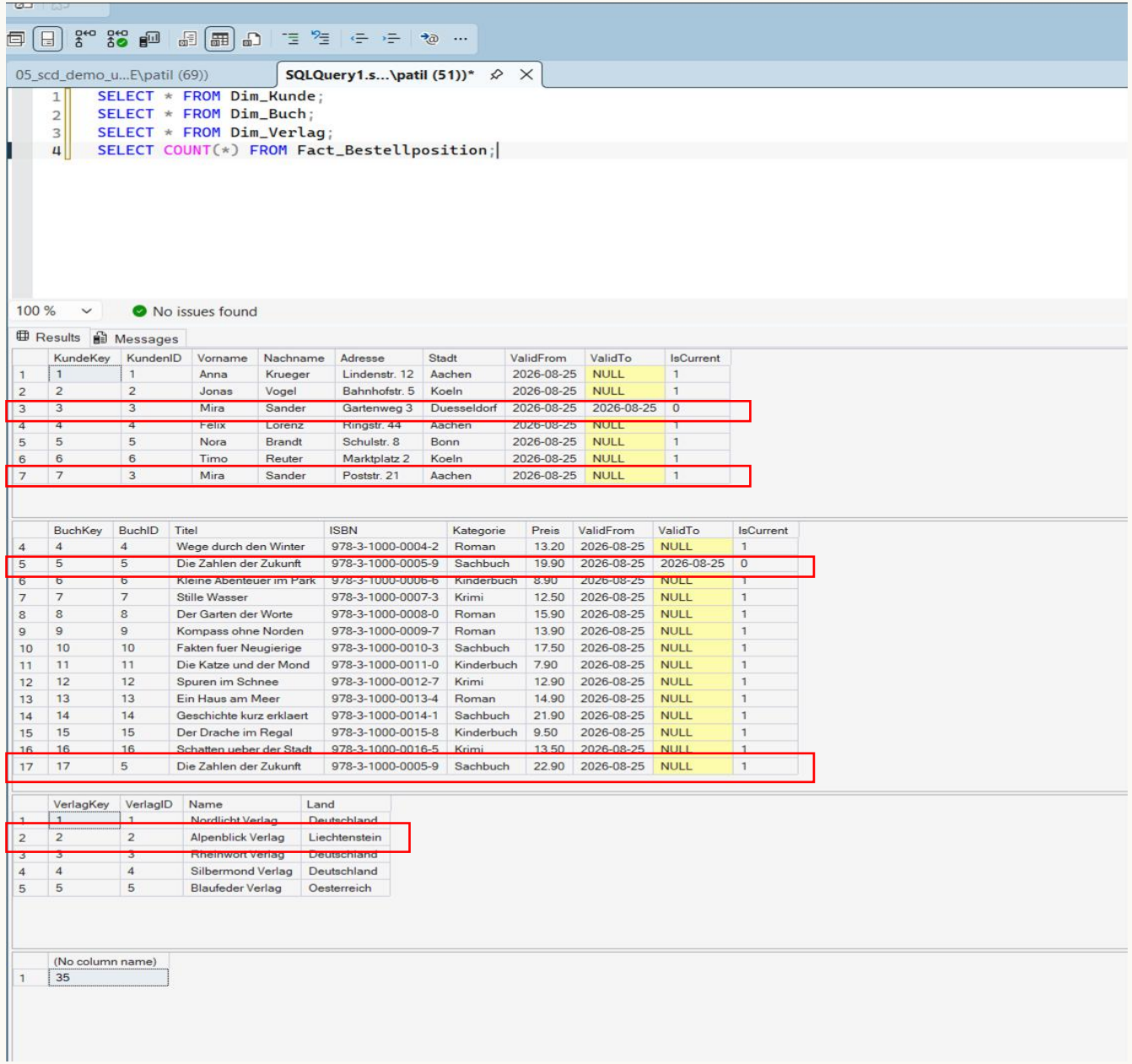
SCD Demonstration

```sql
-- SCD Typ 2 trigger: book price change
UPDATE Buecher SET Preis = 22.90 WHERE BuchID = 5;   -- was 19.90

-- SCD Typ 2 trigger: customer moves
UPDATE Kunden SET Adresse = 'Poststr. 21', Stadt = 'Aachen' WHERE KundenID = 3; -- was Duesseldorf

-- SCD Typ 1 trigger: publisher record correction
UPDATE Verlage SET Land = 'Liechtenstein' WHERE VerlagID = 2;  -- was Schweiz
```





```sql
SELECT * FROM Dim_Kunde;
SELECT * FROM Dim_Buch;
SELECT * FROM Dim_Verlag;
SELECT COUNT(*) FROM Fact_Bestellposition;
```

100 % — No issues found

Results | Messages

	KundeKey	KundenID	Vorname	Nachname	Adresse	Stadt	ValidFrom	ValidTo	IsCurrent
1	1	1	Anna	Krueger	Lindenstr. 12	Aachen	2026-08-25	NULL	1
2	2	2	Jonas	Vogel	Bahnhofstr. 5	Koeln	2026-08-25	NULL	1
3	3	3	Mira	Sander	Gartenweg 3	Duesseldorf	2026-08-25	2026-08-25	0
4	4	4	Felix	Lorenz	Ringstr. 44	Aachen	2026-08-25	NULL	1
5	5	5	Nora	Brandt	Schulstr. 8	Bonn	2026-08-25	NULL	1
6	6	6	Timo	Reuter	Marktplatz 2	Koeln	2026-08-25	NULL	1
7	7	3	Mira	Sander	Poststr. 21	Aachen	2026-08-25	NULL	1

	BuchKey	BuchID	Titel	ISBN	Kategorie	Preis	ValidFrom	ValidTo	IsCurrent
4	4	4	Wege durch den Winter	978-3-1000-0004-2	Roman	13.20	2026-08-25	NULL	1
5	5	5	Die Zahlen der Zukunft	978-3-1000-0005-9	Sachbuch	19.90	2026-08-25	2026-08-25	0
6	6	6	Kleine Abenteuer im Park	978-3-1000-0006-6	Kinderbuch	8.90	2026-08-25	NULL	1
7	7	7	Stille Wasser	978-3-1000-0007-3	Krimi	12.50	2026-08-25	NULL	1
8	8	8	Der Garten der Worte	978-3-1000-0008-0	Roman	15.90	2026-08-25	NULL	1
9	9	9	Kompass ohne Norden	978-3-1000-0009-7	Roman	13.90	2026-08-25	NULL	1
10	10	10	Fakten fuer Neugierige	978-3-1000-0010-3	Sachbuch	17.50	2026-08-25	NULL	1
11	11	11	Die Katze und der Mond	978-3-1000-0011-0	Kinderbuch	7.90	2026-08-25	NULL	1
12	12	12	Spuren im Schnee	978-3-1000-0012-7	Krimi	12.90	2026-08-25	NULL	1
13	13	13	Ein Haus am Meer	978-3-1000-0013-4	Roman	14.90	2026-08-25	NULL	1
14	14	14	Geschichte kurz erklaert	978-3-1000-0014-1	Sachbuch	21.90	2026-08-25	NULL	1
15	15	15	Der Drache im Regal	978-3-1000-0015-8	Kinderbuch	9.50	2026-08-25	NULL	1
16	16	16	Schatten ueber der Stadt	978-3-1000-0016-5	Krimi	13.50	2026-08-25	NULL	1
17	17	5	Die Zahlen der Zukunft	978-3-1000-0005-9	Sachbuch	22.90	2026-08-25	NULL	1

	VerlagKey	VerlagID	Name	Land
1	1	1	Nordlicht Verlag	Deutschland
2	2	2	Alpenblick Verlag	Liechtenstein
3	3	3	Rheinwort Verlag	Deutschland
4	4	4	Silbermond Verlag	Deutschland
5	5	5	Blaufeder Verlag	Oesterreich

	(No column name)
1	35

ETL Schedule

Bookstore Analytics - refresh frequency per dimension/fact

Dim_Kunde	*SCD Type 2*	**Daily (Batch)**	Address changes are rare and not time-critical - daily historization is sufficient.
Dim_Buch	*SCD Type 2*	**Daily (Batch)**	Price changes don't need to be visible in real time.
Dim_Verlag	*SCD Type 1*	**Weekly (Batch)**	Publisher data (name, country) changes very rarely.
Dim_Datum	*static*	**Loaded once, upfront**	Calendar dimension - no ongoing ETL need.
Fact_Bestellposition	-	**Daily after close of business**	No real-time analysis required; a daily batch is sufficient.

What we learned - key takeaways

Highlights

- A data warehouse exists to answer business questions fast, like "revenue per book" , something a normalized Business DB isn't built for.
- SCD is a business decision, not a default - we version history only where it has analytical value (address, price), not fields like email nobody would ask "what was this at the time" about.
- Seeing SCD Type 2 actually work in SQL Server, with two real versions of one customer, made it click far more than the theory did.

Problems + Solutions

- SCD demo showed 0 changes at first, the wrong database was active in SSMS; fixed by switching to the right one.
- Windows setup took longer than expected, python wasn't found and pyodbc was missing; fixed with the py launcher and a pip install.
- Re-running the ETL once nearly created duplicate order rows in the fact table, added a check so it skips anything already loaded, making the script safe to run more than once.

What my colleagues should know

- Always check which database is active before running a script, the easiest mistake to make.
- Decide what needs historical tracking based on business need, not by default.
- Make ETL scripts unchanged, always check if something already exists before inserting it, so re-running is safe.
- Document assumptions like surrogate keys explicitly, even when they seem obvious.

Questions